REVISION TEST -V

CLASS : 12TH STD. **MARKS : 70**

SUBJECT : CHEMISTRY **TIME : 2.30HRS**

PART-I

Chosse the correct answers. **15 X 1 =15**

1. When the dilution increases by 100 times , the dissociation increases by____ time
   a) 100 b) 1 c) 10 d) 25
2. The number of electrons that have a total charge of 9650 coulombs is
   a) 6.22×10^{23} b) 6.022×10^{24} c) 6.022×10^{22} d) 6.022×10^{-34}
3. The phenomenon observed when a beam of light is passed through a colloidal solution is
   a) Cataphoresis b) Electrophoresis c) Coagulation d) Tyndall effect
4. Carbolic acid is
   a) Phenol b) Picric acid c) benzoic acid d) phenylacetic acid
5. Which one of the following reduces tollens reagent
   a) formic acid b) acetic acid
   c) benzophenone d) none of these
6. Use of chloropicrin is as
   a)Explosive. b) dye c) anaesthetic d)sterilizing agent
7. The number of sp2 and sp3 hybridised carbon in fructose are respectively
   a) 1 and 4 b) 4 and 2 c) 5 and 1 d) 1 and 5
8. The metal oxide which cannot be reduced to metal by carbon is
   a) PbO b) Al_2O_3 c) ZnO d) FeO
9. Carbon atoms in fullerene with formula C_{60} have
   a) sp3 hybridised b) sp hybridised
   c) sp2 hybridised. d) partially sp2 and partially sp3 hybridised
10. Which one of the following compounds is not formed?
    a) $XeOF_4$ b) XeO_3 c) XeF_2 d) NeF_2
11. _____form oxo captions
    a) lanthanide b) actinides c) noble gases d) alkali metals
12. The ion with maximum number of unpaired electron is
    a) Mn^{2+} b) Ti^{2+} c) V^{3+} d) Fe^{2+}
13. The sum of primary valency and secondary valency of the metal M in the complex. $[M(en)_2 (Ox)]Cl$
    a)3 b)6 c)-3 d)9

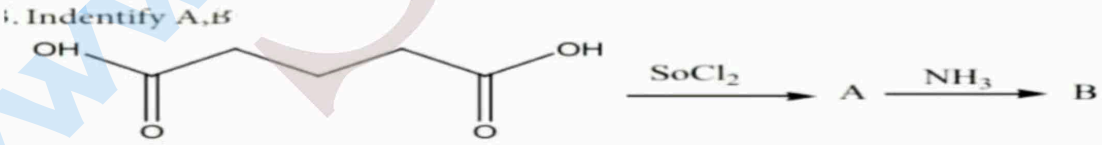
14. The cation leaves its normal position in the crystal and moves to some interstitial position, the defect in the crystal is known as
    a) Schottky defect b) F center
    c) Frenkel defect d) non-stoichiometric defect
15. Assertion: rate of reaction doubles when the concentration of the reactant is doubles if it is a first order reaction.
    Reason . : rate constant also doubles
    a) Both assertion and reason are true and reason is the correct explanation of assertion.
    b) Both assertion and reason are true but reason is not the correct explanation of assertion.
    c) Assertion is true but reason is false.
    d) Both assertion and reason are false.

PART-II

Answer the following any six questions. **6 X 2 = 12**

Question number 24 is compulsory.

16. Give the uses of borax
17. Why do zirconium and hafnium exhibit similar properties ?
18. . Give the difference between double salt and coordination compound .
19. State Bragg's equation.explain it terms
20. Define solubility product.
21. Identify the auto catalyst in the following reaction
    $CH_3COOC_2H_5 + H2O. \rightarrow CH_3COOH + C_2H_5OH$
    $2AsH_3 \rightarrow 2As + 3H_2$
22. What is glacial acetic acid ?
23. how is phenolphthalein prepared ?
24. i. Indentify A,B



PART-III

Answer the following any six questions. **6 X 3 = 18**

Question number 33 is compulsory.

25. which metal in the 3d series exhibits + 1 oxidation State most frequently and why ?
26. $Fe(CN)_6]^{3-}$ paramagnetic , explain using VB theory
27. explain the Deacons's process for manufacture of chlorine
28. A solution of silver nitrate is electrolysed for 20 minutes with a current of 2 ampere calculate the mass of silver deposited at the cathode.

29. What happens when hydrogen sulphide gas is passed through a solution of arsenic oxide name the chemical method.
30. Write the haloform reaction with an example
31. What are reducing and non-reducing sugars ?
32. Write a short note on ultrafiltration
33. Ksp of Ag_2CrO_4 is 1.1 X 10^{-12} .what is the solubility of Ag_2CrO_4 in 0.1M K_2CrO_4

PART-IV

Answer the following all questions. **5 X 5 = 25**

34.a) I)Explain zone refining process with an example

ii) explain why fluorine always exhibit an oxidation state of -1 ?

(OR)

b) I) Write a short note on fullerene

ii) What are the hybridisation of ion in IF_7 ? Give its structure

35 a) I) .Draw the following compounds

I)pyrophospheric acid ii) Hypophosphoric acid

ii) Calculate the number of unpaired electrons in ,Cr^{3+},Cu^{2+} and calculate the spin only magnatic moments?

(OR)

b)I) what are the limitation of VB theory ?

ii) .What are the characteristics of Ionic solids?

36 a) I)Write Arrhenius equation and explain the terms involved

ii) Show that in case of first order reaction , the time required for 99.9% completion is nearly ten times the time required for half completion of the reaction

(OR)

b)I) what are Lewis acid and bases give one example for each

ii) What is electrochemical series ? how is it useful to predict corrosion

37 a)I) .What is the difference between homogeneous and hetrogenenous catalysis ?

ii) Why carbohydrate are generally optical active ?

(OR)

b)How to distinguish 1^0 , 2^0 , and 3^0 alcohol by victor Meyer test ?

38 a) I)Explain the mechanism of cannizzaro reaction .

ii) Write a structure of all possible type updates which can be obtained from glycine and alanine

(OR)

b)name the reducing agent used in the reduction of nitrobenzene to the following compounds

A)Aniline B)phenyl hydroxylamine C)Nitroso benzene
D)azobenzene E)Hydrazobenzene